

Student Exam Score Prediction Using Machine Learning Based on Academic and Behavioral Factors

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ABSTRACT

Student academic performance is influenced by several academic, behavioral, and learning-related factors. Identifying these factors and predicting examination performance before the final assessment can provide useful information for students and educators. This paper presents a machine learning-based Student Exam Score Predictor that estimates the expected examination score of a student using multiple performance-related attributes.

The proposed system considers factors such as study hours, attendance percentage, previous examination marks, assignment performance, participation in academic activities, sleep patterns, and other learning-related characteristics. Historical student records are collected and processed before training the prediction model. Data preprocessing techniques are applied to handle missing values, inconsistent data, and categorical attributes. Feature selection is then performed to identify the most relevant factors associated with examination performance.

Regression-based machine learning algorithms are used to learn the relationship between student characteristics and examination scores. The developed models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2). The trained model can predict the expected examination score of a student based on newly provided academic and behavioral information.

The proposed system can help students understand their expected academic performance and identify areas that may require improvement. It can also assist teachers and educational institutions in identifying students who may need additional academic support. The study demon-

strates the application of machine learning and educational data analytics for proactive academic performance assessment.

1 INTRODUCTION

Student academic performance is an important indicator of the effectiveness of the learning process. Academic performance is usually measured using examination scores, assignments, attendance, classroom participation, and other academic activities. However, the final examination score is affected by multiple factors and cannot always be estimated accurately using a single academic measurement.

Students may have different learning habits, study schedules, attendance levels, previous academic performance, and levels of participation in academic activities. Factors such as the amount of time spent studying, previous examination marks, assignment performance, attendance, sleep patterns, and participation in academic activities can provide useful information about a student's expected performance.

Traditional academic evaluation generally focuses on the marks obtained after an examination has been conducted. Although this approach provides an accurate record of completed performance, it provides limited opportunity for early intervention. A prediction system can provide an estimated performance level before the examination and may help students identify areas that require additional attention.

Machine learning provides an approach for analyzing historical student records and identifying patterns associated with academic performance. Supervised machine learning algorithms can learn the relationship between input features and a target examination score. Since an examination score is a continuous numerical value, regression algorithms are suitable for this problem.

Cortez and Silva demonstrated the application of data mining techniques for analyzing

and predicting secondary school student performance using academic, demographic, social, and school attributes [1]. Educational data mining has also been investigated for analyzing student learning behavior and academic outcomes [2].

This research proposes a Student Exam Score Predictor that uses academic and behavioral information to estimate the expected examination score. The system follows a sequence of data collection, preprocessing, feature selection, machine learning model development, evaluation, and prediction. [?].

2 Literature Review

Machine learning and educational data mining have been widely studied for analyzing student performance.

Cortez and Silva developed a data mining approach for predicting secondary school student performance. Their study investigated academic, demographic, social, and school-related factors and demonstrated the usefulness of data mining for educational performance analysis [1].

Asif et al. investigated undergraduate student performance using educational data mining techniques. Their research showed that historical academic information can be used to analyze and understand student performance patterns [2].

Romero and Ventura presented a survey of Educational Data Mining and Learning Analytics techniques. Their work highlighted the importance of applying data-driven methods to educational data for understanding learning processes and student outcomes [3].

Existing studies demonstrate that machine learning can be applied to student performance prediction. However, student performance is influenced by different combina-

tions of academic and behavioral characteristics. Therefore, the proposed project focuses on combining multiple measurable factors to predict a continuous examination score.

Table 1: Comparison of Existing and Proposed Work

Study	Approach	Focus
Cortez and Silva	Data Mining	Secondary Student Performance
Asif et al.	Educational Data Mining	Undergraduate Performance
Romero and Ventura	Educational Data Mining	Learning Analytics
Proposed Work	Regression ML	Exam Score Prediction

3 Problem Statement

Student examination performance is affected by several academic and behavioral factors. Conventional assessment methods mainly measure performance after an examination has been completed and may not provide an early estimate of a student's expected score.

The problem addressed in this project is to develop a machine learning-based system that can analyze student-related information and predict an expected examination score.

The system uses historical student records containing relevant academic and behavioral factors. A regression model is trained to learn the relationship between these factors and examination scores. The trained model is then used to predict the expected score for a new student.

4 Objectives

The objectives of the proposed Student Exam Score Predictor are:

1. To collect student academic and behavioral information.
2. To preprocess student data before applying machine learning algorithms.
3. To identify important factors affecting examination performance.
4. To select relevant features for score prediction.
5. To develop regression-based machine learning models.
6. To predict the expected examination score of a student.
7. To evaluate the performance of different regression models.
8. To provide useful information for identifying areas where students may require academic improvement.

5 Proposed System

The proposed Student Exam Score Predictor is a machine learning-based system designed to estimate the expected examination score of a student.

The system takes several student-related attributes as input. These attributes may include study hours, attendance percentage, previous examination marks, assignment performance, participation in academic activities, sleep patterns, and other measurable learning-related characteristics.

The collected data is first processed to remove inconsistencies and prepare it for machine learning. Missing values are handled, categorical data is converted into numerical form, and irrelevant features are removed.

Feature selection is then performed to identify the factors that are most relevant to examination score prediction.

The processed dataset is divided into training and testing datasets. Regression algorithms

are trained using the training data. The testing data is used to evaluate the performance of the trained models.

After training, a new student's information can be provided to the system. The trained model analyzes the input features and generates an estimated examination score.

6 System Architecture

The architecture of the proposed Student Exam Score Predictor is shown in Fig. 1.

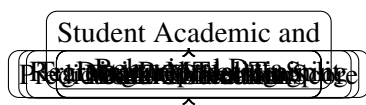


Figure 1: Architecture of the Student Exam Score Predictor

7 Input Features

The proposed system uses academic and behavioral characteristics as input features.

Table 2: Student Input Features

Feature	Description
Study Hours	Average time spent studying per day
Attendance	Percentage of classes attended
Previous Marks	Marks obtained in previous examinations
Assignment Performance	Performance in academic assignments
Academic Participation	Participation in academic activities
Sleep Pattern	Average daily sleep duration
Exam Score	Target examination score

These features are selected because they represent different aspects of student academic

behavior. The actual features used by the implementation depend on the dataset available for training.

8 Dataset

The proposed system requires historical student records containing input characteristics and corresponding examination scores.

Each record represents a student and contains information that can be used to learn the relationship between student behavior and academic performance.

The dataset may contain numerical attributes such as study hours, attendance percentage, previous marks, assignment scores, and sleep duration. Categorical attributes, if present, are converted into numerical representations during preprocessing.

The target variable of the proposed system is the examination score.

If a public student-performance dataset is used, the features and target variable must be reported according to the actual dataset.

9 Data Preprocessing

Data preprocessing is performed before model training to improve the quality and consistency of the dataset.

9.1 Data Cleaning

The dataset is checked for duplicate records, invalid values, missing values, and inconsistent entries.

9.2 Missing Value Handling

Missing numerical values are handled using suitable statistical methods such as mean or median replacement. Categorical missing values can be handled using the most appropriate category-based method.

9.3 Categorical Encoding

Categorical variables are converted into numerical values using encoding techniques such as label encoding or one-hot encoding.

9.4 Feature Scaling

Numerical features may be scaled when required by the selected machine learning algorithm.

9.5 Train-Test Split

The dataset is divided into training and testing subsets. The training subset is used for learning the relationship between student attributes and examination scores. The testing subset is used to evaluate the trained model.

10 Feature Selection

Feature selection is performed to identify the student characteristics that provide useful information for examination score prediction.

Correlation analysis can be used to determine the relationship between numerical input features and examination scores. Tree-based models can also provide feature importance values.

Removing irrelevant features can reduce model complexity and allow the model to focus on the most informative student characteristics.

11 Machine Learning Models

The proposed system uses regression algorithms because the target variable is a continuous examination score.

11.1 Linear Regression

Linear Regression models the relationship between input variables and the examination

score using a linear equation.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \quad (1)$$

where $\hat{y}$ represents the predicted examination score, x_i represents an input feature, β_i represents the corresponding coefficient, and β_0 is the intercept.

11.2 Decision Tree Regression

Decision Tree Regression divides student records into different groups according to feature values and produces a numerical prediction at the leaf nodes.

This method can represent nonlinear relationships between student characteristics and examination performance.

11.3 Random Forest Regression

Random Forest Regression combines multiple decision trees to generate a final prediction.

The final prediction can be expressed as:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x) \quad (2)$$

where N represents the number of trees and $T_i(x)$ represents the prediction generated by the i -th tree.

12 Proposed Methodology

The complete methodology of the Student Exam Score Predictor is divided into the following stages.

12.1 Data Collection

Historical student records containing academic and behavioral information are collected.

12.2 Data Preprocessing

The collected records are cleaned and transformed into a format suitable for machine learning.

12.3 Feature Selection

Relevant student characteristics are identified and selected.

12.4 Model Training

The selected regression algorithms are trained using the training dataset.

12.5 Model Evaluation

The trained models are evaluated using the testing dataset and standard regression metrics.

12.6 Score Prediction

The final trained model receives the information of a new student and generates an expected examination score.

13 Evaluation Metrics

The models are evaluated using Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and R^2 score.

13.1 Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

MAE represents the average absolute difference between actual and predicted examination scores.

13.2 Mean Squared Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

MSE penalizes larger prediction errors more strongly because the errors are squared.

13.3 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

RMSE represents the prediction error in the same unit as the examination score.

13.4 Coefficient of Determination

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

The R^2 score represents the proportion of variation in examination scores explained by the model.

14 Results

The developed machine learning models are evaluated using the testing dataset.

Table 3: Performance of Machine Learning Models

Model	MAE	MSE	RMSE	R^2
Linear Regression	—	—	—	—
Decision Tree	—	—	—	—
Random Forest	—	—	—	—

The values in Table 3 should be replaced with the actual values obtained after executing the machine learning implementation.

15 Model Performance Visualization

A comparison graph can be used to visualize the prediction error of the different regression models.

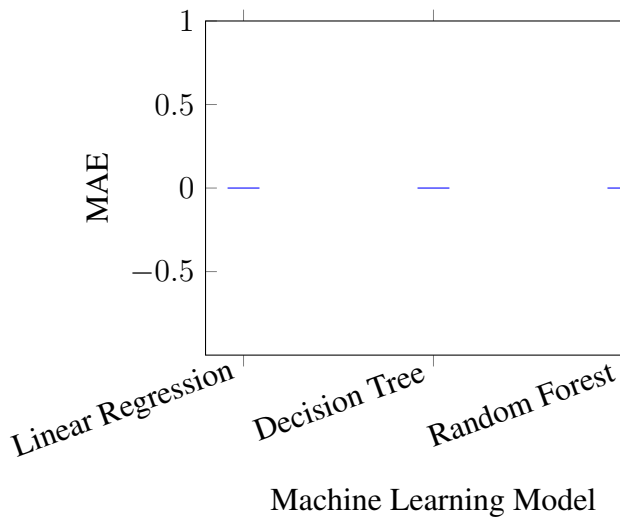


Figure 2: Comparison of model prediction errors

17 Feature Importance Analysis

Feature importance analysis can be used to determine which student-related factors contribute most strongly to the prediction.

Table 4: Feature Importance Analysis

Feature	Importance
Previous Marks	—
Study Hours	—
Attendance	—
Assignment Performance	—
Academic Participation	—
Sleep Pattern	—

16 Actual and Predicted Score Analysis

The actual-versus-predicted graph provides a visual comparison between the examination scores obtained by students and the scores predicted by the trained model.

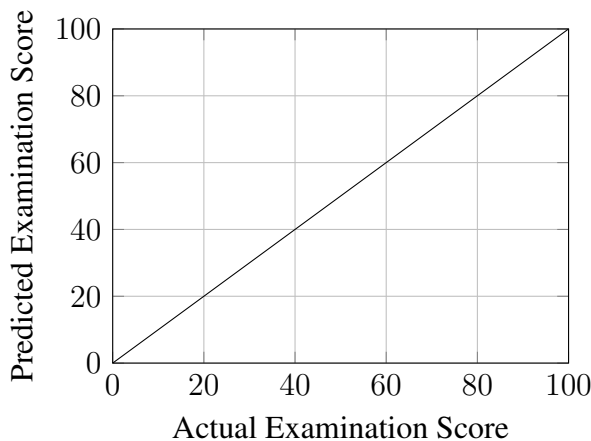


Figure 3: Actual versus predicted examination scores

Points closer to the diagonal reference line indicate smaller differences between actual and predicted examination scores.

The actual feature importance values should be obtained from the trained machine learning model.

18 Discussion

The Student Exam Score Predictor analyzes multiple academic and behavioral characteristics to estimate examination performance.

The use of multiple input features allows the model to consider different aspects of student learning behavior rather than depending only on previous examination marks.

Linear Regression provides a simple baseline for understanding the relationship between student attributes and examination scores. Decision Tree Regression can capture nonlinear relationships between features. Random Forest Regression can combine several decision trees and model more complex relationships.

The experimental results should be used to compare these models using MAE, MSE, RMSE, and R^2 . A lower prediction error indicates smaller differences between predicted and actual examination scores, while the R^2 value indicates how much of the variation

in examination scores is explained by the model.

The prediction generated by the system should be considered an estimated academic indicator rather than a guaranteed examination result.

19 Applications

The proposed Student Exam Score Predictor can be applied in several educational scenarios.

1. **Student Self-Assessment:** Students can use the predicted score to understand their expected performance.
2. **Academic Improvement:** Students can identify areas such as study time, attendance, or previous performance that may require improvement.
3. **Teacher Support:** Teachers can use prediction results as an additional source of information when monitoring student performance.
4. **Early Academic Intervention:** The system can help identify students whose predicted performance indicates a possible need for additional academic support.
5. **Personalized Learning:** Student-specific information can support personalized academic planning.
6. **Educational Analytics:** Institutions can analyze student data to identify patterns in academic performance.

20 Limitations

The prediction accuracy of the proposed system depends on the quality and quantity of the available student data.

Student examination performance can also be affected by factors that may not be represented in the dataset. These factors may in-

clude individual learning preferences, motivation, personal circumstances, examination difficulty, and changes in study behavior.

Therefore, the predicted score should not be interpreted as an exact future examination result. It represents an estimate based on patterns learned from historical student records.

The system may also have limited generalization if the training dataset represents only a specific group of students or a specific educational institution.

21 Future Scope

The proposed system can be extended in several ways.

First, larger and more diverse student datasets can be used to improve the generalization of the prediction model.

Second, additional measurable academic and behavioral factors can be incorporated when reliable data is available. Examples include learning-platform activity, assignment submission patterns, study schedules, and attendance trends.

Third, advanced machine learning and deep learning techniques can be investigated to determine whether they provide improvements over traditional regression algorithms.

The system can also be developed as a web or mobile application. Students could enter their academic information and receive an estimated examination score through an interactive interface.

Explainable machine learning techniques can also be incorporated to show which input factors contributed to an individual prediction.

22 Conclusion

This paper presented a machine learning-based Student Exam Score Predictor for es-

timating examination performance using academic and behavioral factors.

The proposed system processes historical student records, performs feature selection, trains regression models, evaluates their performance, and generates an expected examination score for new student information.

The use of study hours, attendance, previous examination marks, assignment performance, academic participation, sleep patterns, and other relevant characteristics provides a broader representation of student learning behavior.

Regression algorithms such as Linear Regression, Decision Tree Regression, and Random Forest Regression can be used to model the relationship between these factors and examination scores. The models can be evaluated using MAE, MSE, RMSE, and R^2 metrics.

The proposed system can provide useful information for students, teachers, and educational institutions. Students can use the prediction as an indication of their current academic position, while educators can use it as an additional tool for identifying students who may require academic support.

Overall, the project demonstrates how machine learning and educational data analytics can be applied to make student performance assessment more data-driven and proactive.

References

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